

Apollo 17 astronauts captured this image—the celebrated “Blue Marble” photograph—on December 7. Seeing our planet from space helped raise concern for the environment.

## THE LIMITS TO GROWTH

A simple graph (right) illustrates that while population grows geometrically—the more people there are, the faster it grows—a finite resource, such as food supply, can only increase at a steady rate. This inevitably leads to crisis at some point, showing that continued growth cannot be sustained indefinitely.



## THE MICROELECTRONICS

**INDUSTRY** produced a number of advances in 1972. American company Magnavox introduced the Odyssey, the world's **first home video game console**, which plugged into televisions. The popular Pong video game—a two-dimensional on-screen version of table tennis—was released by Atari in November.

The **first digital watch**, the Hamilton Watch Company's Pulsar, went on sale in the fall. The company had announced the watch's development in 1970 and begun producing it in 1971.

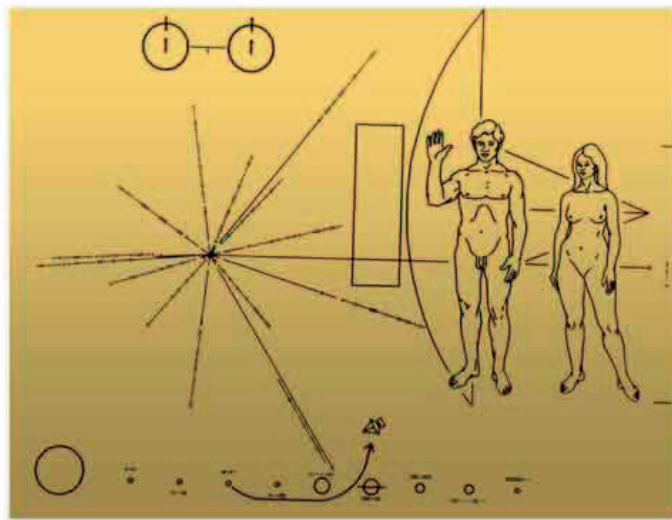
Computer users had been able to send information across the telephone network since the mid-1960s, but in December, Vadic Corporation **introduced the first practical modem**, the VA3400. It could send data at 1,200 bits per second.

In January, the international think tank The Club of Rome published *The Limits to Growth*. It set out the results of a computer simulation investigating the possible effects of **unbridled industrial development and population growth**. The report had many critics at the time, but was nonetheless very important for **raising public awareness of environmental concerns**, such as the affects of economic growth on finite natural resources.

In May, a team led by American geneticist **Walter Fiers** revealed that they had worked out the **sequence of nucleotide bases along the entire length of a**

**Information plaque about Earth** *Pioneer 10* carries this plaque, which aimed to give extraterrestrials an idea of the human form and planet Earth's place in the Solar System.

**gene**—the first time this had been accomplished. Also in May, Japanese-born evolutionary biologist **Susumu Ohno** introduced the term “**junk DNA**.” Geneticists knew that mutations introduced when DNA replicates inside cells impose a limit on the number of genes a genome can carry. That upper limit is around 30,000. Since the amount of DNA in each human cell could carry 3 million genes, the **great majority of our genome must have no function**—hence the term “junk DNA.” Most geneticists prefer to use the term “noncoding DNA”: the DNA in genes carry codes used to build proteins inside cells, but other DNA may yet be found to have other functions.



“IT WAS JUST A ‘EUREKA’. YOU KNOW, I LOOK AT THE **TV SET** AND SAY TO MYSELF, ‘**WHAT CAN I DO WITH THIS?**’ ”

Ralph Baer, inventor of Magnavox Odyssey video game console, 2007

In October, a team of American molecular biochemists led by **Paul Berg** reported that they had **combined sequences of DNA from different organisms**. This process of making “recombinant DNA” is central to genetic engineering.

In July, NASA launched the Earth Resources Technology Satellite—the first of the USA's Landsat satellites. NASA launched seven satellites to **gather information about Earth**

**from space**—providing data on land use, geology, oceans, lakes and rivers, and pollution. The last two Apollo missions, 16 and 17, took another six astronauts to the Moon. Three further Apollo missions had been planned, but were canceled, partly because the budget was cut and partly so that NASA could concentrate on developing an orbiting space station. In March, NASA launched its Pioneer 10 probe, bound for Jupiter (which it flew past in 1974).

**245 POUNDS**  
**THE WEIGHT OF MOON ROCKS COLLECTED BY APOLLO 17 ASTRONAUTS**

